Physics (PHYS-1101)

MECHANICS

Lecture 8:

Thermal Properties of Matter

(Thermal Physics)

Thermal Physics

- Thermal physics is the study of
  - Temperature
  - Heat
  - How these affect matter

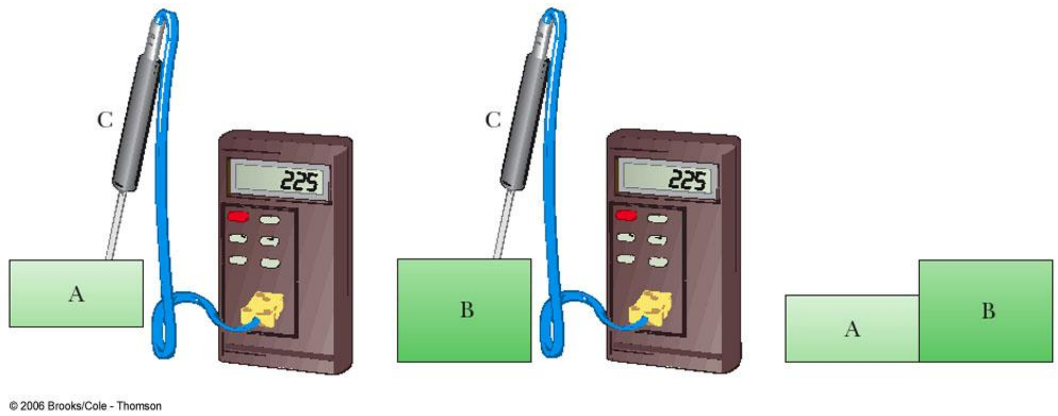
Thermal Physics, cont

- Concerned with the concepts of energy transfers between a system and its environment and the resulting temperature variations.
- Historically, the development of *thermodynamics* paralleled the development of *atomic theory*.
- Concerns itself with the physical and chemical transformations of matter in all of its forms: solid, liquid, and gas.

Heat

- The process by which energy is exchanged between objects because of temperature differences is called *heat.*
- Objects are in *thermal contact* if energy can be exchanged between them.
- *Thermal equilibrium* exists when two objects in thermal contact with each other cease to exchange energy.

Zeroth Law of Thermodynamics



- If objects A and B are separately in thermal equilibrium with a third object, C, then A and B are in thermal equilibrium with each other.
- Allows a definition of temperature

Temperature from the Zeroth Law

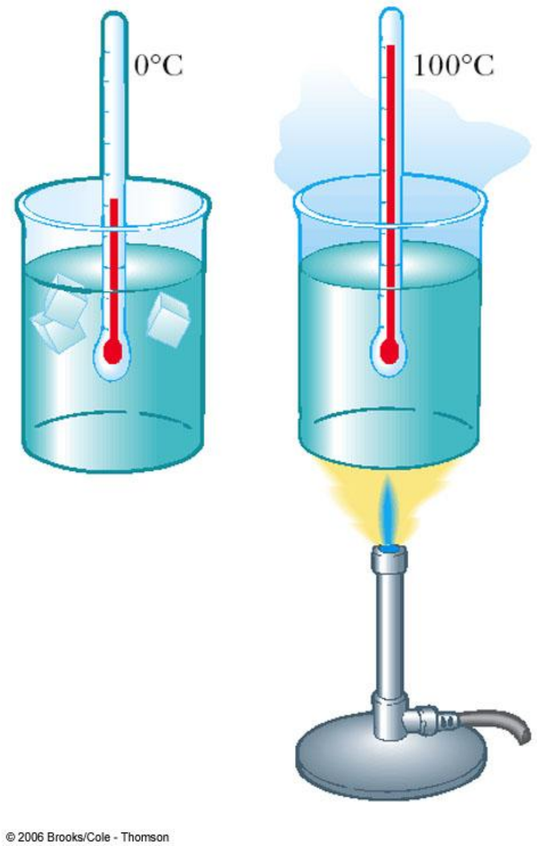
- Two objects in thermal equilibrium with each other are at the *same temperature.*
- *Temperature* is the property that determines whether or not an object is in thermal equilibrium with other objects.

Thermometers

- Used to measure the temperature of an object or a system.
- Make use of physical properties that change with temperature
- Many <u>physical properties</u> can be used
  - *volume* of a liquid
  - *length* of a solid
  - *pressure* of a gas held at constant volume
  - *volume* of a gas held at constant pressure
  - *electric resistance* of a conductor
  - *color* of a very hot object

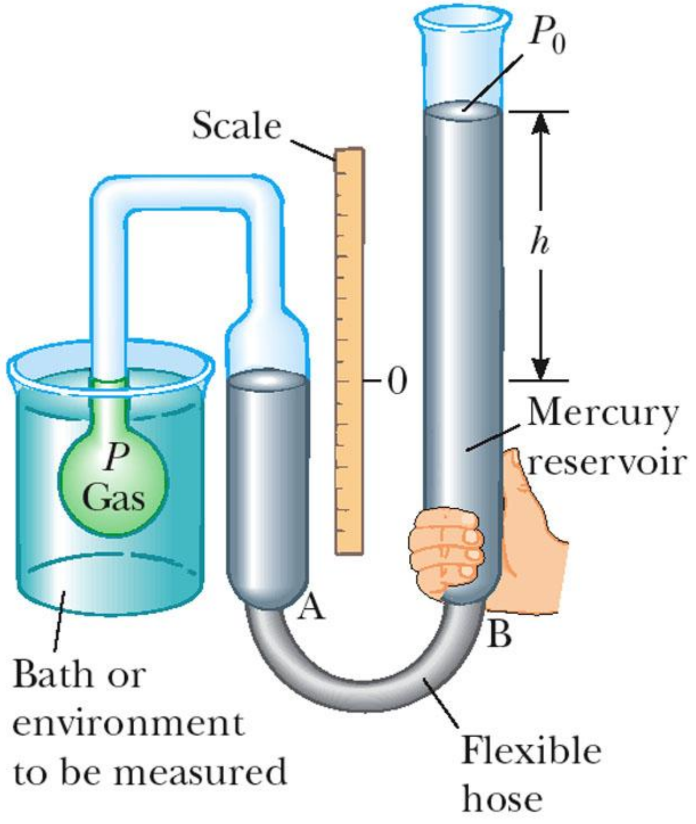
Mercury Thermometers

- A mercury thermometer is an example of a common thermometer
  - The level of the mercury rises due to thermal expansion
  - Temperature can be defined by the height of the mercury column



Gas Thermometer

- Temperature readings are nearly independent of the gas.
- Pressure varies with temperature when maintaining a constant volume



Temperature Scales

- Thermometers can be calibrated by placing them in thermal contact with an environment that remains at constant temperature.
  - Environment could be *mixture of ice and water in thermal equilibrium.*
  - Also commonly used is *water and steam in thermal equilibrium.*

Celsius Scale

- Temperature of an ice-water mixture is defined as 0º C.
  - This is the *freezing point* of water
- Temperature of a water-steam mixture is defined as 100º C.
  - This is the *boiling point* of water
- Distance between these points is divided into 100 segments or degrees

Kelvin Scale

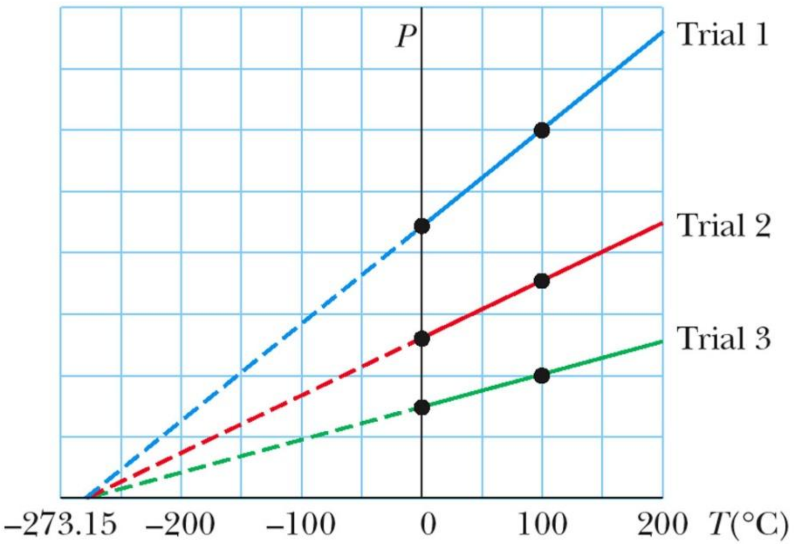
- When the pressure of a gas goes to zero, its temperature is -273.15° C.
- This temperature is called *absolute zero*
- This is the zero point of the Kelvin scale
  - -273.15° C = 0 K
- To convert: $T_C = T_K - 273.15$
  - The size of the degree in the Kelvin scale is the same as the size of a Celsius degree.

Example 1

- A pan of water is heated from 25°C to 80°C. What is the change in its temperature on the Kelvin scale and on the Fahrenheit scale?
  - 55K
  - What about Fahrenheit scale?

Pressure-Temperature Graph

- All gases extrapolate to the same temperature at zero pressure.
- This temperature is *absolute zero.*



Modern Definition of Kelvin Scale

- Defined in terms of two points
  - Agreed upon by International Committee on Weights and Measures in 1954
- First point is absolute zero.
- Second point is the *triple point* of water.
  - Triple point is the single point where water can exist as solid, liquid, and gas.
  - *Single temperature and pressure*
  - Occurs at 0.01º C and P = 4.58 mm Hg

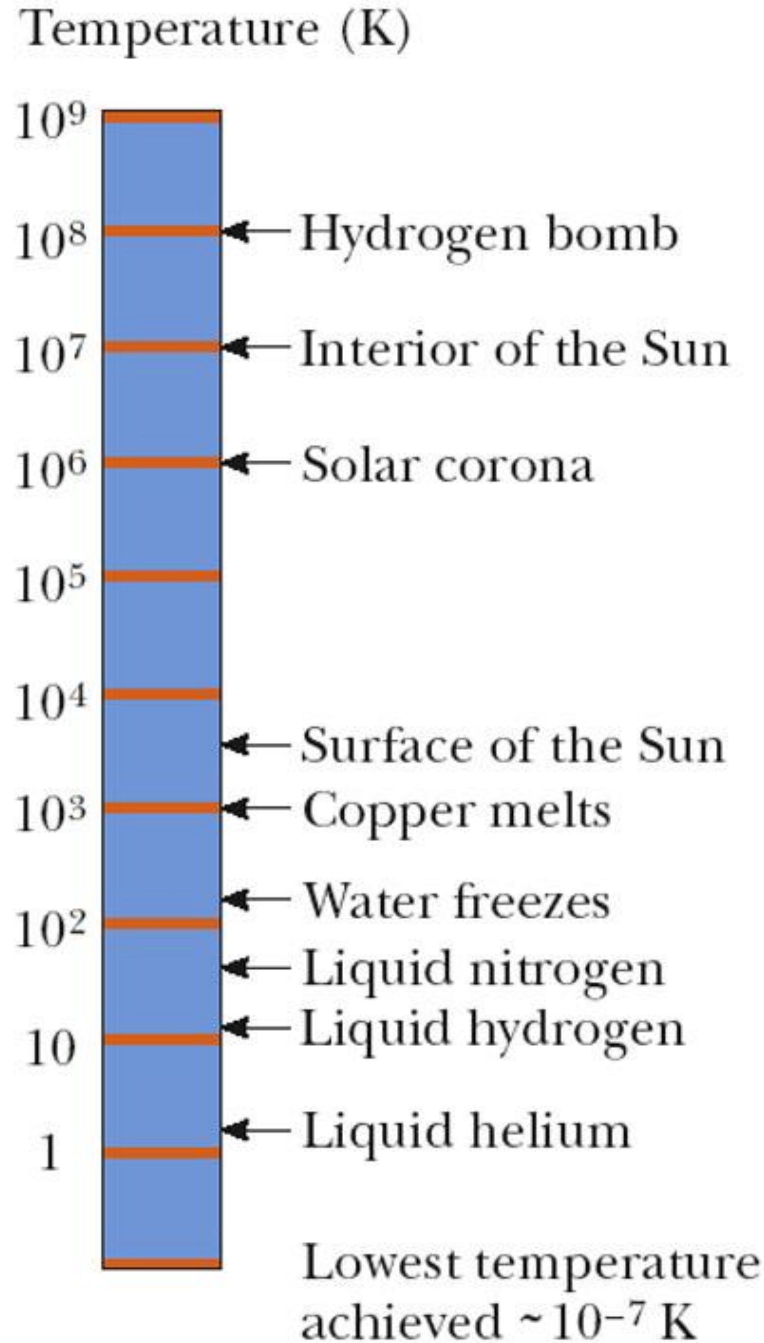
Modern Definition of Kelvin Scale, cont

- The temperature of the triple point on the Kelvin scale is 273.16 K
- Therefore, the **current definition of the Kelvin** is defined as

  1/273.16 of the temperature of the triple point of water.

Some Kelvin Temperatures

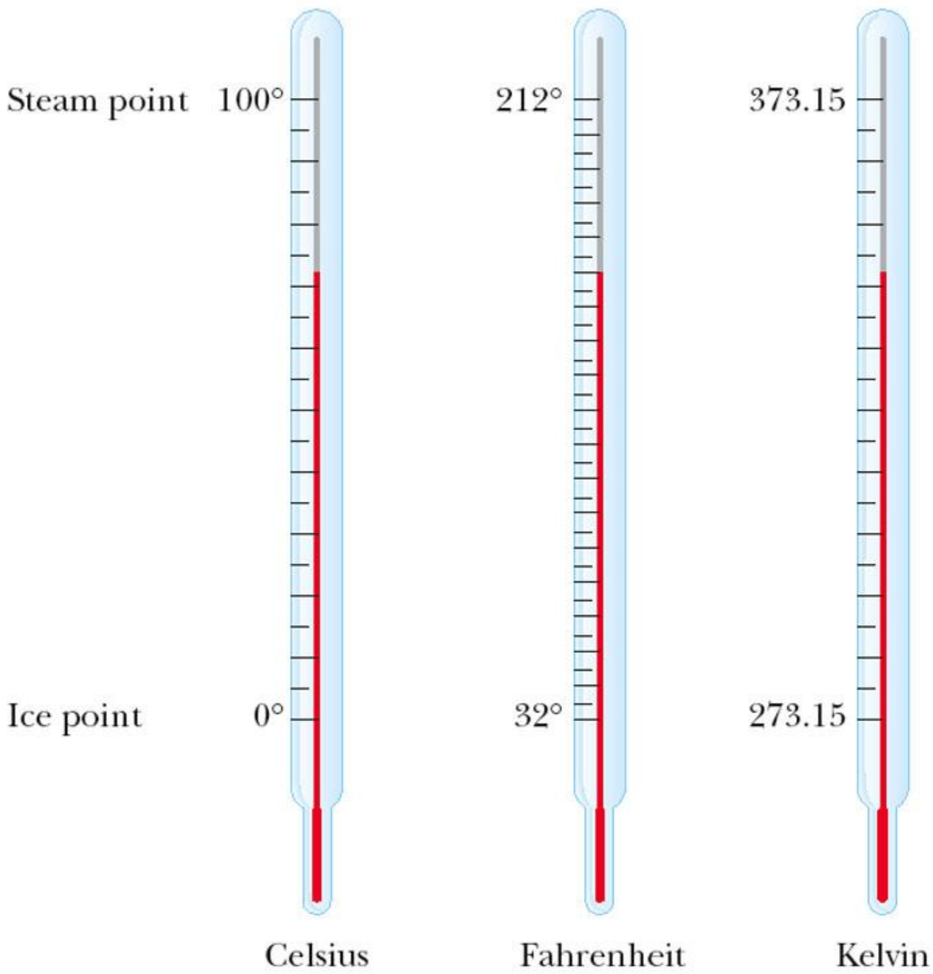
- Some <u>representative Kelvin temperatures</u>
- Note, this scale is logarithmic
- Absolute zero has never been reached!!!



Fahrenheit Scales

- Most common scale used in the US.
- Temperature of the *freezing point* is **32º**.
- Temperature of the *boiling point* is **212º.**
- 180 divisions between the points

Comparing Temperature Scales



Converting Among Temperature Scales

$$T_C = T_K - 273.15$$

$$T_F = \frac{9}{5} T_C + 32$$

$$T_C = \frac{5}{9}\left(T_F - 32\right)$$

$$\Delta T_F = \frac{9}{5} \Delta T_C$$

Exercise

1. An object has the temperature of 50 °F, what is its temperature in degrees Celsius and in Kelvins?
2. A pan of water is heated from 25 °C to 80 °C. What is the change in its temperature on the Kelvin scale and on the Fahrenheit scale? (phy 4 sci & engine pg 586).
3. A spray can containing a propellant gas at twice atmospheric pressure (202 kPa) and having a volume of 125 cm3 is at 22 °C. It is then tossed into an open fire. When the temperature of the gas in the can reaches 195 °C, what is the pressure inside the can? Assume any change in the volume of the can is negligible.

Thermal Expansion

- The thermal expansion of an object is a consequence of the change in the average separation between its constituent atoms or molecules.
- At ordinary temperatures, molecules vibrate with a small amplitude.
- As temperature increases, the amplitude increases.
- This causes the overall object as a whole to expand.

Linear Expansion

- For small changes in temperature

$$\Delta L = \alpha\, L_o\, \Delta T \;\; or \;\; \Delta L = \alpha\, L_o \left(T - T_o\right)$$

,α the coefficient of linear expansion, depends on the material

See table 10.1

- These are average coefficients; they can vary somewhat with temperature.

Applications of Thermal Expansion – Bimetallic Strip

Thermostats

Use a *bimetallic strip*

Two metals expand differently

- Since they have different coefficients of expansion

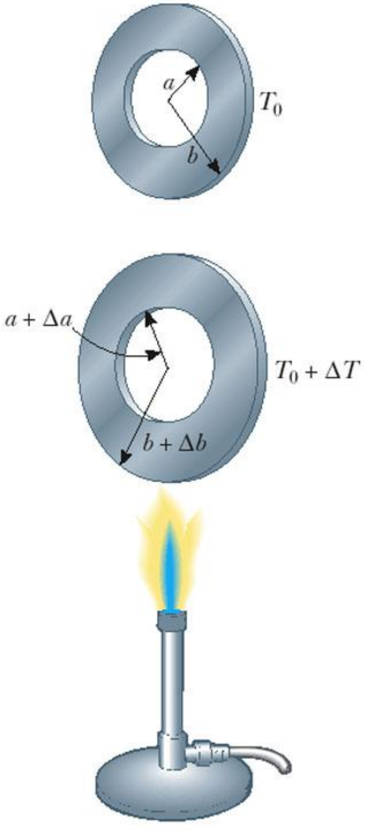
Area Expansion

Two dimensions expand according to

$$\Delta A = \gamma \ A_o \ \Delta t,$$

$$\gamma = 2\alpha$$

▪ is the coefficient of area expansion



Volume Expansion

- Three dimensions expand

$$\Delta V = \beta\, V_o\, \Delta t$$

$$\text{for solids, } \beta = 3\alpha$$

For liquids, the coefficient of volume expansion is given in the table

Exercise

1. A segment of steel railroad track has a length of 30.000 m when the temperature is 0.0°C. What is its length when the temperature is 40.0°C?
2. Show that the change in area of a rectangular plate is given by $\Delta A = 2\alpha Ai\Delta T$.

Thank You For Your Attention !!!